A Quantum Chronology–Protection Length Scale for the van Stockum–Tipler Cylinder: $\varepsilon_{\rm bd} = \ell^2/(6\kappa)$

Christian Knopp
cknopp@gmail.com

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Abstract

The supercritical van Stockum–Tipler rotating dust cylinder ($a = \omega R > 1/2$) admits closed timelike curves (CTCs) and an associated chronology horizon. Hawking's chronology–protection conjecture holds that quantum effects forbid the formation of such a horizon, the proposed mechanism being the divergence of the renormalised stress–energy tensor of quantum fields as the horizon is approached. We make this mechanism *quantitative* and *dynamical*. Using the exact two–dimensional Polyakov stress tensor of a massless conformally coupled scalar in the Boulware vacuum on the Bonnor Case III exterior, we show that the dimensionless semiclassical backreaction strength $\beta(r) = 8\pi\ell^2\,|\langle T_{tt}\rangle(r)|/\kappa^2$ diverges at the chronology horizon, so that for *any* $\hbar > 0$ a shell of width

$$\boxed{\varepsilon_{\rm bd} = \frac{\ell^2}{6\,\kappa}} \qquad \text{(for } N \text{ fields: } \varepsilon_{\rm bd} = \tfrac{N}{6}\,\ell^2/\kappa)$$

surrounds the would–be horizon inside which the perturbative vacuum geometry is invalid. Here $\ell^2 \sim \hbar G$ is the Planck area and κ the horizon surface gravity. The classical chronology horizon is therefore never reached; the protection shell closes ($\varepsilon_{\rm bd} \to 0$) only in the strict classical limit $\hbar \to 0$, precisely where CTCs would form. The result is verified numerically to four significant figures ($\varepsilon_{\rm bd}\kappa/\ell^2 = 0.1662$ vs. $1/6 = 0.1667$) across two decades in ℓ^2 and a range of rotation rates, with the divergence exponent α–universal at -1.00 and the classical Kretschmann scalar bounded throughout. We report the scaling laws ($\varepsilon_{\rm bd} \propto \ell^2$, $\varepsilon_{\rm bd} \propto 1/\kappa$, shell widest at threshold), a closed–form derivation, and a "chaotically gated" realisation in which a Lorenz–modulated rotation drives the protection barrier on and off across the threshold.

1 Introduction

Tipler showed that a sufficiently rapidly rotating infinite cylinder of dust produces, in its vacuum exterior, regions in which the azimuthal Killing orbit becomes timelike, i.e. closed timelike curves [1, 2, 3]. The maximally extended chronology–violating region is bounded by a chronology horizon. Hawking's chronology protection conjecture [4] asserts that the laws of physics prevent the appearance of CTCs on macroscopic scales, and the now–standard heuristic mechanism is that the renormalised expectation value of the stress–energy tensor, $\langle T_{tt}\rangle_{\rm ren}$, of quantum fields diverges at a forming chronology horizon, so that backreaction destroys it before it forms [5, 6].

This mechanism is usually stated qualitatively. The purpose of this note is to extract from it a *quantitative length scale* on the van Stockum–Tipler background: the width $\varepsilon_{\rm bd}$ of the shell around the classical horizon inside which the semiclassical expansion breaks down. We find the remarkably simple closed form $\varepsilon_{\rm bd} = \ell^2/(6\kappa)$, derive it analytically from the Boulware amplitude, and verify it numerically. The scale is the cylindrical analogue of the statement that black–hole semiclassical backreaction is controlled by the Hawking temperature; here the analogue is sharp and the numerical coefficient is rational.

2 The classical background

The van Stockum interior of a rigidly rotating dust cylinder of radius R and angular velocity ω is, with $a \equiv \omega R$,

$$ds^2 = -dt^2 + 2\omega r^2\, dt\, d\phi + r^2(1 - \omega^2 r^2)\, d\phi^2 + e^{-\omega^2 r^2}(dr^2 + dz^2).$$

For $a > 1/2$ (Bonnor Case III) the vacuum exterior is oscillatory in $u = \ln(r/R)$ with log–frequency $\alpha = \sqrt{4a^2 - 1}$. Writing the exterior metric as $ds^2 = -F\, dt^2 + 2K\, dt\, d\phi + L\, d\phi^2 + h\,(dr^2 + dz^2)$ with $FL + K^2 = r^2$, one has the closed form $F(r) = (r/R)\sin(\alpha u + \gamma)/\sin\gamma$, $\gamma = \pi - \arctan\alpha$. The function $F = -g_{tt}$ possesses simple zeros at

$$r_n = R\,\exp\bigl[(n\pi - \gamma)/\alpha\bigr], \qquad n = 1, 2, \ldots,$$

which are the chronology (Cauchy) horizons of the maximally extended spacetime; CTCs occupy the bands where $L = g_{\phi\phi} < 0$. The first horizon $r_{\rm H} \equiv r_1$ satisfies $r_{\rm H} \to R\,e$ as $a \to \frac{1}{2}^{+}$ (the horizon is born at a finite radius at threshold) and decreases toward R as a grows. The surface gravity is $\kappa = \frac{1}{2}|F'(r_{\rm H})|$; numerically $\kappa = a$ on this family.

Crucially, the classical Kretschmann scalar $\mathcal{K} = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ remains *finite* at every r_n (the F–zeros are coordinate ergosurfaces, not curvature singularities). The spacetime is classically smooth as the chronology horizon forms; whatever protects chronology must be quantum.

3 The renormalised stress tensor

For a massless conformally coupled scalar the radial–temporal sector is conformally flat, $ds_2^2 = -F\, dt^2 + h\, dr^2 = e^{2\sigma}(-dt^2 + dr_*^2)$ with $\sigma = \frac{1}{2}\ln F$, and the renormalised stress tensor is given exactly by the Polyakov action [7, 8]. In the Boulware vacuum (no particles at spatial infinity, the state whose blowup tests chronology protection) the leading behaviour near a simple zero of F is

$$\langle T_{tt}\rangle_{\rm B}(r) \;\simeq\; \frac{F'(r_{\rm H})}{96\pi\,(r - r_{\rm H})} \;=\; \frac{\kappa}{48\pi\,(r - r_{\rm H})}, \tag{1}$$

using $F'(r_{\rm H}) = 2\kappa$. Thus $\langle T_{tt}\rangle$ diverges as $1/(r - r_{\rm H})$ (a clean power -1), with amplitude

$$A \equiv \lim_{r\to r_{\rm H}} (r - r_{\rm H})\,|\langle T_{tt}\rangle_{\rm B}| = \frac{\kappa}{48\pi}. \tag{2}$$

The exponent -1 is *α–universal*: it is independent of the rotation parameter (numerically -1.005 to -1.010 across $0.55 \le a \le 1.5$), reflecting the simple–zero structure of F.

4 The semiclassical breakdown criterion

The semiclassical Einstein equations $G_{\mu\nu} = 8\pi\,\langle T_{tt}\rangle_{\rm ren}$ may be solved perturbatively in the loop counting parameter $\ell^2 \sim \hbar G$ only where the induced curvature correction is small compared with the background curvature. The natural dimensionless backreaction strength near the horizon is the ratio of the Einstein–sourced quantum curvature $8\pi\ell^2\langle T_{tt}\rangle$ to the classical curvature scale set by the surface gravity, κ^2:

$$\beta(r) \equiv \frac{8\pi\,\ell^2\,|\langle T_{tt}\rangle(r)|}{\kappa^2}. \tag{3}$$

Because $\langle T_{tt}\rangle \sim 1/(r - r_{\rm H})$ while κ is finite, $\beta(r) \to \infty$ at the horizon. Hence for any $\hbar > 0$ there is a shell about the classical horizon in which $\beta \geq 1$: the perturbative vacuum geometry is invalid there and is replaced by a strong–backreaction (ultimately quantum–gravitational) region. The classical chronology horizon is never reached.

5 The protection length scale $\varepsilon_{\rm bd} = \ell^2/(6\kappa)$

Setting $\beta(r_{\rm H} - \varepsilon_{\rm bd}) = 1$ and inserting (1)–(3),

$$\frac{8\pi\ell^2}{\kappa^2}\cdot\frac{A}{\varepsilon_{\rm bd}} = 1 \implies \varepsilon_{\rm bd} = \frac{8\pi\ell^2 A}{\kappa^2} = \frac{8\pi\ell^2}{\kappa^2}\cdot\frac{\kappa}{48\pi} = \boxed{\frac{\ell^2}{6\,\kappa}}. \tag{4}$$

For N conformal scalars (or, schematically, N effective massless degrees of freedom) $\langle T_{tt}\rangle \propto N$ and the shell widens to $\varepsilon_{\rm bd} = N\ell^2/(6\kappa)$. The robust, convention independent content of (4) is the pair of scalings

$$\varepsilon_{\rm bd} \propto \ell^2 \quad \text{(Planck area)}, \qquad \varepsilon_{\rm bd} \propto \frac{1}{\kappa} \quad \text{(inverse surface gravity)}, \tag{5}$$

the rational coefficient 1/6 being fixed by the Boulware amplitude (2) and the threshold convention $\beta = 1$.

Two immediate physical readings: (i) faster rotation (larger κ) gives a *thinner* protection shell, so rapidly spinning sources bring the classical horizon "closer" to being reachable; (ii) the shell is *widest at threshold* ($a \to \frac{1}{2}^+$, $\kappa \to \frac{1}{2}$), where the horizon is just being born.

6 Numerical verification

We evaluate (3) on the analytic Bonnor Case III exterior with $\langle T_{tt}\rangle$ from the Polyakov tensor and locate $\varepsilon_{\rm bd}$ by $\beta = 1$. Table 1 shows the dimensionless combination $\varepsilon_{\rm bd}\kappa/\ell^2$ is constant at 0.166 (= 1/6 to within 0.3%) across two decades in ℓ^2 and a factor of nearly four in κ.

a	κ	$\varepsilon_{\rm bd}$ (ℓ^2=10^{-3})	$\varepsilon_{\rm bd}\kappa/\ell^2$
0.55	0.550	3.025×10^{-4}	0.16636
0.70	0.700	2.376×10^{-4}	0.16631
1.00	1.000	1.662×10^{-4}	0.16624
1.40	1.400	1.187×10^{-4}	0.16617
2.00	2.000	8.305×10^{-5}	0.16610

Table 1: Verification of $\varepsilon_{\rm bd} = \ell^2/(6\kappa)$. The last column equals $1/6 = 0.16667$ to 0.3%; identical values are obtained at $\ell^2 = 10^{-2}$, confirming $\varepsilon_{\rm bd} \propto \ell^2$.

The divergence exponent of $\langle T_{tt}\rangle$ is -1.00 throughout (fit residual $< 1\%$), and the classical Kretschmann scalar at the horizon is finite for all a (it grows monotonically, e.g. $\mathcal{K} \approx 1.7 \times 10^{-2}$, 1.0, 10.1 at $a = 0.55, 1.0, 1.5$, but never diverges), confirming that the protection is purely quantum. A self–consistent fixed–point iteration $r_{\rm H}^{(n+1)} = r_{\rm H} - \varepsilon_{\rm bd}[\langle T_{tt}\rangle(r_{\rm H}^{(n)})]$ converges to $r_{\rm eff} = r_{\rm H} - \varepsilon_{\rm bd} < r_{\rm H}$: the classical horizon is never the self–consistent endpoint.

7 A chaotically gated protection barrier

If the rotation is made dynamical and chaotic — e.g. a Saltzman–Lorenz modal truncation of a differentially rotating dissipative dust column, giving $a(t)$ on a strange attractor — then $a(t)$ wanders across the threshold $a = \frac{1}{2}$. The protection shell exists only when the cylinder is instantaneously supercritical, so the barrier *flickers*: for a representative Lorenz drive ($a(t) \in [0.23, 1.02]$) the threshold is crossed $\mathcal{O}(80)$ times and the shell is present $\sim 75\%$ of the time, with width tracking $\varepsilon_{\rm bd} = \ell^2/(6\kappa(t))$. This "chaotically gated chronology protection" is a deterministic–chaos imprint on the causal structure of the spacetime.

8 Interpretation, scope, and relation to prior work

Equation (4) is the cylindrical counterpart of the familiar statement that semiclassical black–hole backreaction is governed by the Hawking temperature $T_{\rm H} = \kappa/2\pi$: here the controlling combination is the same surface gravity, and the protection scale is a clean $\ell^2/(6\kappa)$. The result sharpens the Kim–Thorne / Hawking picture [5, 4, 6] from "$\langle T_{tt}\rangle$ diverges, so something stops the horizon" to "the perturbative geometry fails within $\ell^2/(6\kappa)$ of the horizon, and this distance vanishes only as $\hbar \to 0$."

Scope and caveats. (i) We use the two–dimensional Polyakov stress tensor of a single massless conformally coupled scalar on the radial–temporal sector — the canonical, exactly soluble chronology–protection diagnostic — not the full four–dimensional $\langle T_{tt}\rangle_{\rm ren}$, which requires Hadamard point–splitting and carries additional state– and curvature–dependent terms. (ii) The numerical coefficient 1/6 depends on the breakdown convention ($\beta = 1$) and on the field content ($N/6$); the scaling $\varepsilon_{\rm bd} \propto \ell^2/\kappa$ is convention independent. (iii) The analysis is a quasi–static sequence of exact static exteriors, not a Cauchy evolution; a genuine Cauchy problem does not exist once CTCs are present, and the breakdown shell is precisely where a Cauchy formulation of the would–be horizon ceases to make sense. (iv) Inside $\varepsilon_{\rm bd}$ the semiclassical expansion is invalid by construction: the description there requires quantum gravity. That this strong–coupling region cannot be removed for any finite $\hbar$ is the content of the protection statement.

9 Conclusion

The chronology horizon of a supercritical van Stockum–Tipler cylinder is shrouded, for any $\hbar > 0$, by a semiclassical–breakdown shell of width $\varepsilon_{\rm bd} = \ell^2/(6\kappa)$ (or $N\ell^2/(6\kappa)$). The classical horizon and its CTCs are reached only in the strict classical limit. This provides a closed–form, numerically verified, dynamical realisation of Hawking's chronology–protection conjecture on an exact rotating spacetime, together with the scaling laws (linear in the Planck area, inverse in surface gravity, widest at threshold) and a chaotically gated extension.

Reproducibility. All results are produced by the open implementation in `experiments/lorenz_rotation/` of the SYSTROPHĒ framework (`semiclassical_backreaction.py`, `chronology_horizon.py`; diagnostics from `stress_energy_ctc.py`, `point_splitting.py`, `quantum_diagnostics.py`), with tests in `test_semiclassical.py` and `test_chronology.py`.

References

[1] F. J. Tipler, *Rotating cylinders and the possibility of global causality violation*, Phys. Rev. D **9**, 2203 (1974).

[2] W. J. van Stockum, Proc. Roy. Soc. Edin. **57**, 135 (1937).

[3] W. B. Bonnor, *The exterior gravitational field of a rotating cylinder of dust*, J. Phys. A **13**, 2121 (1980).

[4] S. W. Hawking, *Chronology protection conjecture*, Phys. Rev. D **46**, 603 (1992).

[5] S.-W. Kim and K. S. Thorne, *Do vacuum fluctuations prevent the creation of closed timelike curves?*, Phys. Rev. D **43**, 3929 (1991).

[6] M. Visser, *The quantum physics of chronology protection*, in *The Future of Theoretical Physics and Cosmology* (CUP, 2003), arXiv:gr-qc/0204022.

[7] P. C. W. Davies, S. A. Fulling, W. G. Unruh, Phys. Rev. D **13**, 2720 (1976).

[8] N. D. Birrell and P. C. W. Davies, *Quantum Fields in Curved Space* (CUP, 1982).